

Rigidity of the Black–Scholes Valuation Operator

Representative-Invariant Local Valuation under Fixed One-Factor Market Data

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Word count: 8,433

Abstract

This paper proves a rigidity theorem for the Black–Scholes local valuation operator. Standard derivations begin with a stochastic model and use no-arbitrage replication to obtain the pricing PDE. Here the order of explanation is reversed. The question is which deterministic one-factor local valuation operators can assign a pointwise value to a terminal claim independently of payoff-preserving, self-financing redescrptions of that claim. The operative condition is quotient well-definedness, stated as representative invariance. Within the class of payoff-linear, deterministic, one-state, local, continuous hedge-neutral, self-financing, scale-covariant, time-homogeneous valuation systems with a fixed risk-free numeraire, the only nondegenerate generator is the Black–Scholes operator. The proof is a normal-form exhaustion: any same-scope alternative must alter payoff identity, introduce an auxiliary selector, enlarge the state space, change financing, modify local continuation, break scale or time symmetry, add terminal or boundary data, or change the numeraire. Each route is bookkeeping, a declared enlargement, or a failure of representative invariance. Local volatility, stochastic volatility, jumps, stochastic rates, funding frictions, and incomplete-market measure selections are controlled scope changes, not counterexamples. This supplies a necessity result complementary to stochastic sufficiency derivations while preserving empirical model plurality within finance. The result clarifies stochastic representations as coordinates of a fixed valuation operator.

Keywords. Black–Scholes equation; valuation operators; mathematical finance; parabolic PDE; representative invariance; option pricing.

1 Introduction

The Black–Scholes equation is usually introduced through a stochastic model and a self-financing replication argument. This presentation is sufficient for deriving the familiar pricing PDE, but it leaves a separate operator-classification question: which local valuation generators remain possible when the value is required to be invariant under payoff-preserving self-financing redescrptions of the same claim? This paper answers that question for the fixed one-factor local interface. The result is not an empirical assertion that markets are lognormal, nor a proposal for a new pricing model. It is a rigidity theorem for the local valuation operator itself.

Scope statement. All uniqueness and non-existence claims in this paper are comparison-class relative. The theorem does not assert that empirical option markets satisfy the fixed class, nor that

richer models are illegitimate. It asserts that, after the listed identity, locality, continuity, symmetry, financing, and numeraire data are held fixed, no non-Black–Scholes local operator remains available without changing the valuation problem. Departures from Black–Scholes identify precisely which structural condition is violated or deliberately relaxed. These data are used as a valuation interface, not as a stochastic model of empirical markets.

This distinction matters because Black–Scholes continues to function as a valuation coordinate even when the literal stochastic assumptions are relaxed. Market prices are quoted through implied volatility; empirical departures are recorded as smiles, skews, local-volatility surfaces, stochastic-volatility factors, jumps, stochastic rates, collateral conventions, and funding adjustments rather than as a wholesale abandonment of the Black–Scholes coordinate. A minimal working understanding of standard option-pricing PDE methods is sufficient to follow the argument.

The paper separates two questions that are usually run together.

1. Which stochastic, numerical, or martingale representation is convenient for computing a price?
2. Which local valuation operator assigns a value to the claim itself, independently of payoff-preserving self-financing redescriptions of that claim?

The second question is prior to the first at the level of operator classification. If the numerical value changes when the same terminal claim is rewritten through a lawful self-financing hedge convention, coordinate convention, or equivalent representative, then the number is not yet a value of the claim alone. It is a value of the claim plus a selected representation. The question addressed here is therefore not which stochastic model reproduces observed prices, but which local operator assigns a value invariant under all representation-preserving transformations of the same claim. Any alternative notion of valuation that relaxes this invariance defines a different valuation object.

Contribution and relation to standard derivations

The main result is a rigidity theorem for local valuation generators. Standard Black–Scholes–Merton arguments show that a particular stochastic-replication representation is sufficient to obtain the Black–Scholes PDE. The present paper proves the converse kind of statement inside a fixed local valuation interface: once payoff linearity, deterministic pointwise valuation, self-financing invariance, continuous hedge-neutral local continuation, scale covariance, time homogeneity, terminal trace, and a fixed risk-free numeraire are imposed, the only nondegenerate representative-invariant local operator is the constant-coefficient Black–Scholes operator.

The role of the hypotheses is analogous to familiar PDE classification arguments in which locality, regularity, symmetry, and normalization data determine the admissible operator form. Translation invariance and locality, for example, restrict possible differential generators before any particular solution is studied; parabolicity and terminal data then determine the relevant evolution problem. Here the financial inputs play the same structural role: payoff linearity, self-financing invariance, scale covariance, time homogeneity, and the fixed cash numeraire restrict the admissible local valuation generator before any stochastic representation or calibration procedure is selected. The Black–Scholes operator is therefore obtained as the normal form of the fixed valuation interface, not as an empirical modeling preference.

The theorem is conditional, but the conditions are not ornamental or reverse engineered. Each condition records a piece of structure required for the phrase ‘the local value of this claim’ to be independent of the representative used to describe the claim. Dropping any condition admits

familiar non-Black–Scholes valuation families; restoring it blocks those families as same-scope rivals. Thus the paper does not claim that markets are lognormal, that volatility is empirically constant, or that richer models are illegitimate. It classifies what remains after the valuation problem is held fixed.

The argument is organized as a normal-form exhaustion rather than as an empirical model derivation. We first formalize claim identity as a quotient of descriptions by payoff-preserving self-financing redescription. We then isolate the data fixed by this quotient and the data that must be imported from outside it. A same-scope rival to Black–Scholes can depart only through finitely many loci: the quotient, auxiliary data, state variables, financing relation, local continuation law, symmetry class, terminal trace, or numeraire. The proof closes each locus. What remains is a linear second-order local generator. Scale covariance gives the S^2 , S , and constant coefficient degrees; time homogeneity makes them constants; the cash and stock benchmark claims fixed by the risk-free numeraire give the drift and discount coefficients; the only residual nondegenerate scalar is σ^2 .

The counterexample burden is therefore precise. A same-scope counterexample must produce a non-Black–Scholes operator while preserving all of the following simultaneously: the same terminal-payoff quotient, the same one-dimensional local state, the same self-financing redescription relation, deterministic pointwise valuation, continuous hedge-neutral local closure, scale covariance, time homogeneity, the same terminal trace, and the same deterministic risk-free numeraire. A useful richer model does not meet this burden; it identifies which fixed-scope condition has been relaxed. The result lies at the interface of mathematical finance and PDE classification, and is intended as a structural contribution rather than an empirical model. Section 2 defines representative invariance, Section 3 specifies the valuation interface, Sections 4–6 establish operator rigidity, and Section 7 proves exhaustion. Accordingly, the result is not a reformulation of Black–Scholes but a classification theorem: within the stated interface, Black–Scholes is the only operator that defines a representation-independent value of a contingent claim.

Table 1. Core terminology used in the paper.

Term	Use in this paper
Representative invariance	Quotient well-definedness: equivalent descriptions of the same claim receive the same pointwise value.
Admissible redescription	A payoff-preserving, self-financing, selector-free change of description or coordinate; used here as a technical shorthand.
Local state	The declared sufficient state for the fixed problem, here the pair (S, t) or equivalently (S, τ) .
Self-financing	A capital-neutral rebalancing convention that changes implementation but not claim identity.
Valuation operator	The local generator or semigroup assigning values to payoff quotients within the fixed interface.

2 Claim Identity and Representative Invariance

Fix a maturity T , a strictly positive traded underlying level $S > 0$, and a deterministic cash account

$$B_t = e^{rt}, \tag{2.1}$$

where r is constant unless explicitly stated otherwise. The body treats the no-dividend case. Appendix E records the continuous-dividend modification.

Let Φ be a European terminal payoff on $(0, \infty)$. For regularity, the main theorem may be read first on the test class $C_c^2((0, \infty))$ and then extended to standard payoff classes by the usual approximation and boundary-condition arguments. Nothing in the classification depends on the particular dense test class; the test class is used only to make the local generator statements unambiguous.

Definition 2.1 (Description class). *For a fixed terminal payoff Φ , let $\mathcal{D}_\Phi$ denote the class of admissible descriptions of the claim. A description may include payoff notation, portfolio-accounting convention, hedge convention, coordinate normalization, and self-financing representative, provided that it purports to describe the same European terminal claim.*

Definition 2.2 (Admissible redescription). *Let $\mathcal{A}_\Phi$ be the collection of transformations of descriptions satisfying all four clauses:*

1. *they preserve the terminal payoff Φ ;*
2. *they are lawful coordinate changes or are implementable by self-financing rebalancing with no external capital injection;*
3. *they introduce no hidden state variable, calibration criterion, external information source, probability-law selector, extra payoff component, or unobserved convention that can affect value;*
4. *they preserve the economic identity of the original claim rather than replacing it by a richer claim.*

Two descriptions $R, R' \in \mathcal{D}_\Phi$ are admissibly equivalent, written $R \sim R'$, when they are connected by a finite chain of transformations in $\mathcal{A}_\Phi$ and their inverses. The quotient $\mathcal{D}_\Phi / \sim$ is the claim identity at the declared level of description.

Lemma 2.3 (Maximal admissible equivalence closure). *Relative to the declared valuation interface, $\sim$ is the maximal equivalence relation generated by payoff-preserving, self-financing or lawful-coordinate transformations that introduce no valuation-bearing external selector. There is no further free choice of claim identity inside the fixed problem.*

Proof. By construction, $\sim$ contains every finite composition of transformations in $\mathcal{A}_\Phi$ and their inverses, and it contains no pair whose connection requires leaving $\mathcal{A}_\Phi$. A larger relation, in the sense of fewer equivalence classes and more identified pairs, would identify at least one pair of descriptions not connected by admissible transformations. Such an identification must either change the terminal payoff, use external capital, import a selector, or replace the claim by a richer object, so it violates the declared identity interface. A smaller relation, in the sense of more equivalence classes and fewer identified pairs, would fail to identify at least one lawful self-financing or coordinate redescription of the same claim. Then implementation or notation could become valuation-bearing without being detected by the quotient, so the resulting object would not be the claim identity intended here. Thus changing $\sim$ is not a harmless convention; it changes the valuation problem. $\square$

Definition 2.4 (Representative-invariant value). *A valuation assignment $\mathcal{V}$ has representative invariance on $\mathcal{D}_\Phi$ if*

$$R \sim R' \implies \mathcal{V}(R; S, t) = \mathcal{V}(R'; S, t) \quad (2.2)$$

for every admissible pre-maturity state (S, t) . Equivalently, the value is representative-invariant on the claim quotient.

Proposition 2.5 (Quotient factorization). *A valuation assignment has representative invariance if and only if it factors through the claim quotient. That is, there is a map $\bar{\mathcal{V}}$ such that*

$$\mathcal{V}(R; S, t) = \bar{\mathcal{V}}([R]; S, t), \quad [R] \in \mathcal{D}_\Phi / \sim .$$

Proof. If $\mathcal{V}$ factors through the quotient, equivalent descriptions have the same equivalence class and hence the same value. Conversely, if equivalent descriptions always receive the same value, define $\bar{\mathcal{V}}([R]; S, t) = \mathcal{V}(R; S, t)$. This definition is independent of the representative exactly because (2.2) holds. $\square$

Representative invariance is not imposed to select a particular operator. It is not an additional economic preference. It is the minimal condition under which the phrase ‘the value of the claim’ is well-defined independently of its representation. Without it, a valuation may still be a useful value of a model, convention, or richer market specification, but it is not a representative-independent value of the original claim.

Lemma 2.6 (Self-financing necessity from representative invariance). *If two descriptions differ only by a capital-neutral rebalancing convention that preserves the terminal payoff, then a representative-invariant valuation must assign them the same value. Thus self-financing invariance is not an optional microstructure hypothesis inside this comparison class; it is the identity-preserving rule for admissible trading redescrptions.*

Proof. A self-financing rebalance changes holdings without adding external wealth or adding a new payoff component. If the terminal payoff and declared state are unchanged, the rebalance is a different implementation of the same claim rather than a new claim. If a valuation changes under such a rebalance, the value depends on the implementation path or accounting convention rather than on the payoff quotient, contradicting Proposition 2.5. If the rebalance uses external cash, collateral, transaction costs, or funding rules, it is no longer capital-neutral and those cash flows are additional market or payoff data. That case may define a richer valuation problem, but it is not a same-scope redescription. $\square$

Definition 2.7 (Interface-fixed and imported data). *A datum used by a valuation rule is interface-fixed if every admissible redescription carries it to the same datum up to already licensed normalization. A datum is imported if it is supplied by a selector, calibration surface, probability law, hidden state, path history, boundary condition, preference functional, funding convention, or representative choice not fixed by the payoff quotient, the local state, the self-financing relation, and the fixed numeraire.*

Lemma 2.8 (Selector non-invariance). *Any valuation-bearing datum that is not invariant under all admissible redescrptions of the fixed interface cannot be used by a representative-invariant same-scope rule. In particular, an equivalent-martingale-measure choice in an incomplete market, a fitted volatility surface, a calibration loss, or a representative-selection convention is not quotient-fixed merely because it is mathematically convenient.*

Proof. Let d be such a datum. Since d is not invariant under admissible redescription, there are representatives $R \sim R'$ for which the payoff quotient, local state, self-financing relation, terminal trace, and numeraire agree, while d assigns different valuation-bearing information. If the value changes with d , then the value is a function of $([R], d)$ rather than of $[R]$ alone. Proposition 2.5 is

violated. If the value does not change with d , then d is bookkeeping. Declaring d to be part of the market specification may be legitimate, but it changes the input of the valuation problem.

For equivalent martingale measures this says the following. In a complete one-factor Black–Scholes market, replication makes all admissible representations agree on the same price, so the measure is a coordinate representation of an already fixed value. In an incomplete market, distinct equivalent martingale measures generally give different discounted expectations for the same payoff. Choosing one of them is therefore additional market or preference data, not a datum fixed by the original payoff quotient. Calibration surfaces and loss functions are analogous: unless they are declared as market primitives, they select among representatives rather than being fixed by the claim identity. $\square$

Lemma 2.9 (Selector elimination). *A representative-invariant valuation rule cannot depend valuation-essentially on imported data. If an imported datum affects value, then either it must be promoted into the state or market specification, changing the comparison class, or representative invariance fails.*

Proof. This is the contrapositive of Lemma 2.8. If a datum is not fixed by the payoff quotient, local state, self-financing relation, terminal trace, and numeraire, then it is not preserved by the admissible-redescription relation of the fixed problem. Valuation-essential dependence on it therefore prevents quotient factorization. If the datum is added to the declared state or market primitives, the rule may be coherent in the enlarged market, but it is no longer a same-scope valuation rule for the original one-state problem. If no value can change, the datum is bookkeeping. $\square$

Lemma 2.10 (No post-selection repair). *A valuation rule that fails representative invariance cannot be repaired inside the same scope by later averaging, conditioning, calibration, representative selection, or change of measure. Such an operation either defines a new quotient-respecting rule, changes the scope, or leaves the original representative dependence intact.*

Proof. Suppose $\mathcal{V}(R; S, t) \neq \mathcal{V}(R'; S, t)$ for some $R \sim R'$. A later operation P can map the two values to a single number only by ignoring the difference or by using additional information. If it ignores the difference, the difference was not valuation-bearing and was bookkeeping. If it uses a probability law, calibration criterion, representative, conditioning event, or state enlargement, that information is an imported datum unless it was already interface-fixed. Lemma 2.9 then classifies the operation as a scope change or a representative-invariance failure. Therefore same-scope post-selection cannot repair a representative-dependent rule. $\square$

Lemma 2.11 (Expectation agreement is not value identity). *Expectation-level equality does not imply representative invariance. If $R \sim R'$ but $\mathcal{V}(R; S, t) \neq \mathcal{V}(R'; S, t)$ while $\mathbb{E}[\mathcal{V}(R; S, t)] = \mathbb{E}[\mathcal{V}(R'; S, t)]$ under a selected law, then the valuation is representative-indexed and the selected expectation only masks the failure.*

Proof. Representative invariance is equality of the assigned pointwise value on the quotient. Equality after applying an expectation operator is equality after first choosing a probability law and an averaging functional. That choice is invariant under admissible redescription only if the underlying valuation functional already factors through $\mathcal{D}_\Phi/\sim$ or if replication has made the choice immaterial. Otherwise the expectation is itself a selector: two representatives can disagree pointwise while the chosen averaging procedure erases the disagreement. Such erasure does not define a value of the quotient; it defines a value of the quotient together with the averaging rule. Therefore expectation agreement cannot repair representative dependence inside the fixed scope. $\square$

3 The Fixed Local Valuation Interface

This section states the comparison class. The conditions are deliberately explicit so that useful extensions are not misdescribed as counterexamples. These conditions are not restrictive assumptions but the minimal interface required for a pointwise valuation to be well-defined independently of representation. Each item is required to make a pointwise one-factor value independent of representative choice, or else it names the exact interface component whose relaxation creates a different valuation problem.

Definition 3.1 (Local valuation system). *A local one-factor valuation system is a family of operators*

$$\mathcal{P}_\tau : \Phi \mapsto u^\Phi(\cdot, \tau), \quad \tau = T - t \geq 0,$$

where $u^\Phi(S, \tau) = V^\Phi(S, T - \tau)$ is the value at underlying level S and time-to-maturity τ . The terminal trace is

$$\mathcal{P}_0\Phi = \Phi. \tag{3.1}$$

A system is same-scope for this paper only if it preserves the terminal-payoff quotient, the one-dimensional local state S , the deterministic cash account (2.1), and the self-financing redescription relation.

Lemma 3.2 (Locality from state sufficiency). *For a one-factor local valuation system, the statement that value is a function of (S, τ) is a state-sufficiency claim. If the continuation value at fixed (S, τ) depends on another valuation-bearing datum, then that datum is redundant, must be added to the state or market specification, or is an imported selector.*

Proof. Suppose two continuations have the same payoff quotient and the same local state (S, τ) but differ in a datum d . If both receive the same value for all admissible payoffs, d is redundant and the value descends to $V(S, \tau)$. If d changes value and is declared as part of the problem, the state space or market interface has been enlarged. If d changes value while remaining outside the state, Lemma 2.9 classifies it as imported data. Thus locality is not the denial that richer states exist; it is the assertion that (S, τ) is the complete state for the fixed one-factor problem. $\square$

Definition 3.3 (Regular representative-invariant valuation class). *A local valuation system belongs to the fixed representative-invariant class if it satisfies the following conditions.*

- (C1) **Payoff quotient.** *The object valued is $\mathcal{D}_\Phi/\sim$, not a representative in $\mathcal{D}_\Phi$.*
- (C2) **Linearity.** *$\mathcal{P}_\tau$ is linear in terminal payoffs on a regular test class.*
- (C3) **One-factor locality.** *Before maturity, value depends only on (S, τ) and the payoff quotient, not on unrecorded histories, hidden factors, regimes, or path statistics.*
- (C4) **Pointwise determinacy.** *Once interface-fixed inputs are fixed, $u^\Phi(S, \tau)$ is a deterministic pointwise value.*
- (C5) **Self-financing invariance.** *Self-financing rebalancing conventions that introduce no external capital do not change value.*
- (C6) **Terminal-trace fixation.** *The only terminal data for the European problem are the payoff trace (3.1) and the benchmark traces fixed by the cash and stock claims. Additional boundary or selection data are not silently available.*

- (C7) **Continuous hedge-neutral local closure.** *The same-state continuation law is local, continuous, nondegenerate, and stable under self-financing neutralization of first-order stock exposure. The fixed class contains no jump kernel, memory kernel, rough driver, extra local jet, or additional boundary datum.*
- (C8) **Scale covariance.** *Rescaling the monetary unit of the underlying, $S \mapsto \lambda S$ with $\lambda > 0$, changes coordinates but not the valuation law.*
- (C9) **Time homogeneity.** *Calendar shifts preserving time-to-maturity do not change the local valuation law.*
- (C10) **Fixed risk-free numeraire.** *Discounting is with respect to the fixed deterministic cash account (2.1).*

Remark 3.4 (How to read the class). *Condition (C7) is the substantive local-continuation interface. It does not deny jumps, memory kernels, rough drivers, deterministic transport-only valuation, or higher-jet laws. It says that those objects are not part of the fixed one-factor continuous hedge-neutral problem. The second-order form used later is derived from this interface in Lemma 5.1. Conditions (C8)–(C10) then rigidify the coefficients of the remaining operator.*

Lemma 3.5 (Linearity from split-bundle invariance). *In the frictionless self-financing class, payoff linearity is forced by representative invariance for bundled and split descriptions of the same terminal portfolio payoff.*

Proof. The claim paying $\Phi_1 + \Phi_2$ can be described either as one bundled claim or as a self-financing portfolio holding the two claims separately. These descriptions have the same terminal payoff and differ only by admissible portfolio decomposition, so representative invariance requires

$$V^{\Phi_1 + \Phi_2}(S, t) = V^{\Phi_1}(S, t) + V^{\Phi_2}(S, t).$$

Likewise, holding a units of a claim gives payoff $a\Phi$ and value aV^Φ in the frictionless linear-holdings regime. Failure of additivity or homogeneity is possible only when funding costs, inventory constraints, preferences, transaction costs, capital charges, or nonlinear aggregation rules become valuation-bearing. Those are legitimate extensions but not same-scope rivals. $\square$

Lemma 3.6 (Terminal-trace fixation). *Within the fixed European problem, a valuation rule may not use additional terminal, lateral, or calibration boundary data unless those data are already interface-fixed by the payoff quotient and fixed numeraire. If such data affect value, their use is a scope change or a representative-invariance failure.*

Proof. The value problem is declared by the terminal payoff quotient, the local state, the self-financing relation, and the cash numeraire. A further terminal condition, lateral boundary convention, calibration surface, representative selection, or terminal projection criterion is therefore not part of the same input unless it is invariant under all admissible redescriptions and fixed by the declared market interface. If it affects value while not fixed, Lemma 2.9 applies. If it is declared as additional market data, the problem has been enlarged. If it has no effect, it is bookkeeping. $\square$

Proposition 3.7 (Minimality as valuation-interface necessity). *The conditions in Definition 3.3 are necessary for the stated object of study: a deterministic pointwise value of a European payoff quotient on a one-factor local frictionless market interface. Removing any condition does not leave a weaker proof of the same object; it changes the object whose value is being assigned.*

Proof. The payoff quotient is necessary because otherwise the value is not a value of the claim identity; Lemma 2.3 fixes the quotient as the maximal admissible identity relation. Linearity and self-financing invariance are necessary because split and bundled frictionless portfolios, and capital-neutral rebalancings, are admissible redescrptions of the same terminal portfolio payoff; see Lemmas 3.5 and 2.6. One-factor locality is necessary because a value depending on variance, rates, regimes, histories, or funding variables is a value on an enlarged state, as Lemma 3.2 records. Pointwise determinacy is necessary because unresolved randomness leaves no single value at fixed (S, t) . Terminal-trace fixation is necessary because a European terminal-value problem is not determined by the payoff quotient once extra boundary or calibration data are allowed. Continuous hedge-neutral local closure is necessary for the declared nondegenerate continuous one-state local interface; jumps, memory, rough drivers, and higher jets are different continuation laws. Scale covariance and time homogeneity are necessary to prevent the numerical unit of S or the calendar label from becoming a hidden coefficient selector. The fixed numeraire is necessary because changing the discounting asset changes the market interface unless the change is merely a fully specified coordinate redescription. Thus each condition is a component of representative-independent local valuation, not an assumption added solely to obtain a familiar PDE. $\square$

Proposition 3.8 (Strict dependence and necessity ledger). *Each condition in Definition 3.3 is load-bearing in two senses. First, Proposition 3.7 identifies why it is required by the valuation interface. Second, dropping it while retaining the others admits a familiar non-Black–Scholes family; restoring it blocks that family as a same-scope rival.*

Table 2. Dependence and necessity ledger.

Condition removed	Non-Black–Scholes witness	Why it is not same-scope when the condition is restored
Payoff quotient	Representative-indexed prices for the same terminal payoff	Value depends on description rather than claim identity.
Linearity	Utility-indifference, capital-charge, or nonlinear funding valuations	Split and bundled self-financing portfolios no longer receive additive values.
One-factor locality	Stochastic volatility, stochastic rates, regimes, path-dependent statistics	Extra state variables carry valuation information.
Pointwise determinacy	Random valuation functionals or model-uncertainty selectors	Expectation or calibration masks a representative-indexed value.
Self-financing invariance	External funding, collateral, inventory, transaction-cost models	Rebalancing implementation becomes valuation-bearing.
Terminal-trace fixation	Extra boundary conditions, calibration surfaces, representative projections	Boundary or calibration data are imported unless declared as new market data.
Continuous hedge-neutral closure	Jump integro-PDEs, memory kernels, rough drivers, higher local jets, first-order transport	The continuation law is no longer the same nondegenerate local diffusion law.
Scale covariance	Level-dependent coefficients or local-volatility surfaces	The numerical level of S becomes a coefficient selector.
Time homogeneity	Time-dependent volatility or calendar regimes	Calendar time becomes valuation-bearing.
Fixed risk-free numeraire	Stochastic rates, collateral numeraires, funding bases	The discounting asset or numeraire is additional market state.

Proof. The second column lists standard valuation families that become available once the named condition is removed. The third column identifies the datum through which the family enters. Restoring the removed condition either renders the datum inert, promotes it into the market/state

description, or makes the rule representative-dependent. Thus each condition is strictly relevant to the fixed-class conclusion. $\square$

4 Exclusion Mechanisms: No Hidden Selector, No Silent Scope Change

The next lemmas make the exhaustion proof non-enumerative. They show that apparent alternatives are controlled by where their valuation-bearing information enters.

Definition 4.1 (Bookkeeping equivalence). *Two valuation rules are bookkeeping-equivalent on the fixed class if they differ only by admissible redescription, notation, coordinate normalization, or computational representation and induce the same pointwise value on every payoff quotient.*

Definition 4.2 (Same-scope rival). *A candidate is a same-scope rival only if it preserves every condition in Definition 3.3. If it adds a state variable, kernel, volatility surface, stochastic-rate factor, path statistic, nonlinear funding rule, calibration criterion, or numeraire, then it may be a useful model but it is not a same-scope rival.*

Lemma 4.3 (No silent redescription). *A change of variables, measure, numeraire, hedge convention, or calibration convention is an admissible redescription only if it preserves the claim quotient and all fixed-class data. If it changes the state, adds a selector, changes the continuation law, or changes the numeraire, it is a scope change.*

Proof. Admissible redescription is defined by payoff preservation, self-financing implementability, and absence of valuation-bearing imported data. A transformation that supplies a new probability law, volatility surface, state variable, kernel, funding rule, or numeraire changes the data on which value depends. By Proposition 2.5, that move cannot be counted as a different representative of the same quotient unless it has no valuation effect. $\square$

Lemma 4.4 (Asymmetry-source normal form). *Any valuation-bearing asymmetry between two otherwise admissibly equivalent descriptions has exactly one of the following sources:*

- (A1) *a quotient-fixed invariant of the claim and fixed market interface;*
- (A2) *a declared state or market variable, which changes the comparison class if it was absent before;*
- (A3) *an auxiliary selector or representative convention not fixed by the quotient;*
- (A4) *a pure redescription with no valuation effect.*

Only (A1) and (A4) are same-scope; (A2) is a declared scope change; (A3) violates representative invariance if it affects value.

Proof. Let a be the datum responsible for the asymmetry. If a is invariant under all admissible redescriptions and belongs to the declared payoff/state/numeraire/financing interface, then it is (A1). If the rule declares a to be a new market or state component, then it has enlarged the comparison class, giving (A2). If a affects value but is not invariant and not declared, it is imported, giving (A3) and contradicting Lemma 2.9. If it does not affect value, it is bookkeeping, giving (A4). These cases exhaust the possibilities for a valuation-bearing datum. $\square$

Lemma 4.5 (Nonlocal and hidden-state elimination). *A valuation rule whose continuation depends on an integral kernel, jump measure, convolution term, hidden factor, path statistic, volatility surface, stochastic-rate state, funding state, or regime variable is not a same-scope rival unless that datum is redundant.*

Proof. Each listed object supplies valuation-bearing information not contained in the fixed payoff quotient, the local state (S, τ) , the self-financing relation, and the fixed numeraire. If the object affects value while remaining outside the state, Lemma 2.9 excludes it. If it is promoted into the state or market primitives, Definition 3.3 has been changed. If it has no valuation effect, it is bookkeeping. $\square$

Lemma 4.6 (Path-state dichotomy). *Path dependence is compatible with representative invariance only in one of two ways: the relevant path statistic is explicitly included in the state, or the statistic is redundant for value. In the first case the model is no longer same-scope; in the second it descends back to a function of (S, τ) .*

Proof. Take two admissible histories reaching the same local state (S, τ) and the same terminal-payoff quotient. If a path statistic assigns different continuation values to the two histories, then the statistic is valuation-bearing information not contained in the one-factor local state. Leaving it outside the state violates Lemma 2.9; adding it to the state is a declared state enlargement. If all histories with the same (S, τ) and payoff quotient receive the same value, the path statistic has no valuation effect and the valuation factors through $V(S, \tau)$. $\square$

Lemma 4.7 (Nonlinear-valuation elimination). *A nonlinear valuation functional is not a same-scope rival in the frictionless self-financing class.*

Proof. Lemma 3.5 forces additivity and homogeneity for bundled and split descriptions of the same terminal portfolio payoff. A nonlinear rule can survive only by making preferences, funding constraints, inventory, transaction costs, capital charges, or another aggregation rule valuation-bearing. Those data are not part of the fixed quotient. Promoting them changes the class; leaving them hidden violates Lemma 2.9. $\square$

Proposition 4.8 (Three exclusion mechanisms). *Inside the fixed interface, every attempted non-Black-Scholes departure is excluded by one of three mechanisms:*

- (G1) no internal generator: *a new coefficient, measure, kernel, surface, or asymmetry cannot be generated by redescription of the same payoff quotient;*
- (G2) no hidden carrier: *valuation-bearing state, path, rate, volatility, funding, or boundary information must be carried by the declared state or market data;*
- (G3) no post hoc repair: *averaging, calibration, limiting, conditioning, or representative selection cannot restore quotient factorization after it has failed.*

Proof. Mechanism (G1) is Lemmas 2.8 and 4.4: a valuation-bearing distinction is either quotient-fixed, declared as new data, or imported. Mechanism (G2) is Lemmas 4.5 and 4.6: if information affects value, it needs a carrier in the state or market interface. Mechanism (G3) is Lemmas 2.10 and 2.11: a later operation can only ignore the failure or add a selector. The three mechanisms are therefore not additional assumptions; they are the operational forms of quotient factorization. $\square$

Proposition 4.9 (Finite departure locus). *Every non-bookkeeping departure from the fixed local valuation problem must occur at one or more of the following loci:*

- (L1) *the claim-identity quotient;*
- (L2) *an auxiliary selector or imported datum;*
- (L3) *the local state space;*
- (L4) *the self-financing or payoff-linearity relation;*
- (L5) *the terminal trace or boundary data;*
- (L6) *the local continuation law;*
- (L7) *scale covariance, time homogeneity, or the fixed numeraire.*

No eighth same-scope valuation-bearing locus remains.

Proof. A local valuation rule can depend only on the identity of the object being valued, auxiliary data used to select a value, the state variables at which value is evaluated, the portfolio/financing relation that licenses redescription, the boundary/terminal data of the value problem, the continuation operator that propagates the terminal trace, and the symmetries/numeraire that fix coefficient interpretation. These are precisely (L1)–(L7). A purported additional locus is either a new way of specifying one of these items or has no effect on value. $\square$

5 From the Fixed Interface to the Local Generator

We now classify the operator left by the preceding exclusion mechanisms. Work in time-to-maturity $\tau = T - t$, and write

$$u(S, \tau) = \mathcal{P}_\tau \Phi(S).$$

A backward equation in calendar time is recovered by $\partial_t V = -\partial_\tau u$.

Lemma 5.1 (Second-order necessity from hedge neutralization). *Within the continuous one-factor self-financing interface, a nondegenerate local continuation term that survives delta neutralization must appear at curvature order. A first-order term is hedge exposure or deterministic transport; a higher-order term requires additional local jets, nonlocal data, or boundary information not carried by the fixed state.*

Proof. For a smooth value $f(S)$, a local move of the underlying gives

$$f(S + \Delta S) - f(S) = f'(S)\Delta S + \frac{1}{2}f''(S)(\Delta S)^2 + o((\Delta S)^2).$$

The $f'(S)\Delta S$ term is precisely the stock exposure neutralized by holding $f'(S)$ shares in a self-financing hedge. A same-scope residual continuation term cannot depend on an unhedged first-order exposure, because that would make the hedge convention valuation-bearing. If no residual term remains, the rule is deterministic first-order transport and is degenerate relative to the present class. The first nondegenerate hedge-neutral local term is therefore the quadratic curvature term. Terms depending on third or higher derivatives require a stable rule for higher local jets or higher infinitesimal cumulants; jump and memory terms require kernels or histories. None is present in the one-factor local interface, so each is a scope change or imported data by Lemmas 2.9 and 4.5. $\square$

Lemma 5.2 (Operator normal-form completeness). *Any homogeneous linear local generator on scalar C^2 test functions whose valuation-bearing dependence is exhausted by $f(S)$, $f'(S)$, and $f''(S)$ has the form*

$$\mathcal{G}f(S) = A(S, \tau)f''(S) + B(S, \tau)f'(S) - C(S, \tau)f(S).$$

No additional local operator term is available without becoming an inhomogeneous source, a hidden coefficient, a higher jet, or a nonlocal/state-dependent datum.

Proof. Linearity and payoff homogeneity imply that the zero payoff has zero value, so an additive source term independent of f is excluded. Locality in a scalar state means the operator can use only the finite jet of f at S ; Lemma 5.1 restricts the same-scope nondegenerate continuous hedge-neutral dependence to orders at most two. The coefficient multiplying each jet component is a deterministic function of the declared state and time-to-maturity by pointwise determinacy. Any dependence on further variables, kernels, random terms, higher derivatives, or boundary data has already been classified as a state enlargement, continuation-law change, or selector. The displayed normal form therefore exhausts the local operator space left by the interface. $\square$

Lemma 5.3 (Linear local generator ceiling). *A nondegenerate valuation rule in the fixed representative-invariant class of Definition 3.3 has, on the regular test class, a linear second-order local generator. Thus*

$$\partial_\tau u = \mathcal{G}u, \quad \mathcal{G}f(S) = A(S, \tau)f''(S) + B(S, \tau)f'(S) - C(S, \tau)f(S), \quad (5.1)$$

with $A(S, \tau) > 0$ in the nondegenerate case.

Proof. Linearity follows from Lemma 3.5. One-factor locality and Lemma 3.2 exclude dependence on states other than S and on path histories. Pointwise determinacy excludes unresolved stochastic terms in the value. Lemmas 4.5 and 4.7 exclude nonlocal, hidden-state, and nonlinear alternatives inside the fixed scope. Lemma 5.1 shows that, under the continuous hedge-neutral local interface, the residual non-bookkeeping local uncertainty enters through the second derivative of a C^2 value function. Lemma 5.2 then gives (5.1). Nondegeneracy is $A > 0$. $\square$

Lemma 5.4 (Transport, zero-order, higher-order, elliptic, and hyperbolic exits). *A purely first-order transport rule, a zero-order rule, a higher-order local rule, an elliptic boundary-value rule, or a hyperbolic temporal rule is not a nondegenerate same-scope rival.*

Proof. A first-order rule depends on a characteristic transport coefficient. If that coefficient is not fixed by the payoff quotient, self-financing relation, and numeraire, it is imported. If it is fixed to deterministic risk-free transport, the model has no residual local uncertainty and leaves the nondegenerate continuous hedge-neutral class. A zero-order rule cannot propagate arbitrary terminal payoff dependence through the underlying state. A higher-order rule requires higher local jets or additional boundary data not present in the fixed state. Elliptic equations require boundary data rather than a terminal-value propagation law. Hyperbolic equations propagate along characteristics and require characteristic data. Each case either changes the continuation law or imports data excluded by Lemma 2.9. $\square$

Proposition 5.5 (Terminal-value uniqueness inside the generator class). *Within the regular parabolic class (5.1), once the coefficients and terminal trace are fixed, the pointwise valuation is unique on the test class.*

Proof. Let u_1 and u_2 solve the same linear parabolic terminal-value problem with the same coefficients and terminal trace. Their difference $w = u_1 - u_2$ solves the homogeneous problem with zero terminal trace. The standard maximum principle, or equivalently uniqueness for the corresponding strongly continuous positive linear semigroup on the test class, gives $w = 0$. Hence no second value can be obtained without changing coefficients, boundary data, state, or functional class. $\square$

This proposition is included to block a common evasion. A second price for the same terminal trace cannot be obtained by simply choosing a different solution of the same regular local problem. The only possible source of a different value is a change in the data already listed in Proposition 4.9.

6 Coefficient Rigidity

The previous section leaves a linear second-order local operator. The remaining assumptions fix its coefficient form.

Lemma 6.1 (Scale covariance). *Scale covariance forces*

$$A(S, \tau) = \frac{1}{2}a(\tau)S^2, \quad B(S, \tau) = b(\tau)S, \quad C(S, \tau) = c(\tau) \quad (6.1)$$

for scalar functions a, b, c .

Proof. Let the same value be written in the rescaled coordinate $\tilde{S} = \lambda S$, and define $\tilde{f}(S) = f(\lambda S)$. Then

$$\tilde{f}'(S) = \lambda f'(\lambda S), \quad \tilde{f}''(S) = \lambda^2 f''(\lambda S).$$

For the same valuation law to hold in every monetary unit, the coefficients must satisfy

$$A(\lambda S, \tau) = \lambda^2 A(S, \tau), \quad B(\lambda S, \tau) = \lambda B(S, \tau), \quad C(\lambda S, \tau) = C(S, \tau).$$

The locally bounded solutions on $(0, \infty)$ are $A(S, \tau) = \alpha(\tau)S^2$, $B(S, \tau) = b(\tau)S$, and $C(S, \tau) = c(\tau)$. Writing $\alpha = a/2$ gives (6.1). $\square$

Lemma 6.2 (Time homogeneity). *Time homogeneity forces $a(\tau)$, $b(\tau)$, and $c(\tau)$ to be constants.*

Proof. Time homogeneity states that calendar shifts preserving time-to-maturity do not change the local law, and that horizon decomposition uses the same generator over equal elapsed intervals. A deterministic coefficient schedule would make the valuation law depend on a calendar or maturity-specific datum not fixed by the payoff quotient. Time inhomogeneity is admissible only by introducing calendar-dependent structure not fixed by payoff identity. Such schedules are legitimate in a time-inhomogeneous class, but not in the fixed class. Hence a, b, c are constants. $\square$

Lemma 6.3 (Numeraire invariance versus numeraire fixing). *Lawful numeraire changes are admissible coordinate redescrptions only when the conversion rule is itself interface-fixed and carries benchmark claims covariantly. A pointwise PDE representation must nevertheless fix a numeraire; otherwise the drift and discount coefficients are defined only relative to an unspecified accounting coordinate.*

Proof. A change from one numeraire to another is harmless at the quotient level when the relative numeraire process and the induced transformation of values are part of the declared market interface. Then the change is a coordinate redescription of the same value. If the relative numeraire is stochastic, collateral-dependent, funding-dependent, or otherwise not fixed by the original one-factor state and cash account, it supplies additional state or market data. If it is chosen ad hoc, it is a selector. Thus numeraire invariance holds at the claim-identity level, while numeraire fixing is required at the pointwise representation level to prevent the coordinate itself from becoming a hidden valuation-bearing input. In this paper that fixed coordinate is the deterministic cash account (2.1). $\square$

Lemma 6.4 (Risk-free numeraire fixing). *Self-financing invariance with the fixed no-dividend risk-free numeraire gives*

$$b = r, \quad c = r. \quad (6.2)$$

Proof. The claim paying one unit of cash at maturity has value $u(S, \tau) = e^{-r\tau}$. Substituting into $\partial_\tau u = \mathcal{G}u$ gives

$$-re^{-r\tau} = -ce^{-r\tau},$$

so $c = r$. The claim paying the underlying itself at maturity has value $u(S, \tau) = S$ in the no-dividend self-financing class, since it is replicated by holding one share. Substituting $u(S, \tau) = S$ gives

$$0 = bS - cS,$$

so $b = c = r$. $\square$

Corollary 6.5 (Residual curvature scalar). *After quotient invariance, locality, hedge-neutral continuous closure, scale covariance, time homogeneity, terminal trace, and numeraire fixing are imposed, the only remaining nondegenerate coefficient is one positive scalar $a = \sigma^2$.*

Proof. Scale covariance restricts the second-order coefficient to $\frac{1}{2}aS^2$; time homogeneity removes calendar dependence; numeraire fixing determines the first-order and zero-order coefficients. No other coefficient can vary without breaking one of these constraints or importing a selector. Nondegeneracy gives $a > 0$. Writing $a = \sigma^2$, volatility is therefore not an additional same-scope modeling choice; it is the unique residual scalar intensity of hedge-neutral curvature left by the exhaustion. The theory removes all structural degrees of freedom except a single scalar. Empirical calibration of that scalar is an inference problem, not a theoretical indeterminacy or a remaining degree of theoretical freedom inside the fixed class. $\square$

7 Main Necessity and Exhaustion Theorems

Theorem 7.1 (Black–Scholes operator rigidity). *Let $V^\Phi(S, t)$ be a nondegenerate valuation rule in the fixed one-factor representative-invariant local class of Definition 3.3. Then its calendar-time valuation operator is uniquely*

$$\partial_t V + \frac{1}{2}\sigma^2 S^2 \partial_{SS} V + rS \partial_S V - rV = 0, \quad \sigma > 0. \quad (7.1)$$

Proof. Assume, for contradiction, that a nondegenerate same-scope representative-invariant valuation operator exists in Definition 3.3 and is not of the form (7.1). Proposition 2.5 first makes value a function on the claim quotient, and Lemma 2.3 fixes that quotient as the maximal admissible identity relation. Lemmas 2.6 and 3.2 identify self-financing invariance and locality as representation-invariance requirements. Lemmas 2.8–2.11 exclude hidden selectors, post-selection repair, and expectation-only identity. Lemmas 4.3–4.8 exclude silent scope changes, hidden carriers, nonlocal kernels, path statistics, and nonlinear valuation inside the fixed class. Lemmas 5.1, 5.2, and 5.3 reduce the same-scope local continuation law to a linear second-order generator. Lemma 5.4 records why other temporal or local-order types are outside the nondegenerate class. Lemmas 6.1, 6.2, 6.3, and 6.4 force $A = \frac{1}{2}\sigma^2 S^2$, $B = rS$, and $C = r$ in time-to-maturity coordinates. Rewriting $\partial_\tau u = \mathcal{G}u$ as $\partial_t V + \mathcal{G}V = 0$ gives exactly (7.1), contradicting the assumed non-Black–Scholes form. Hence the operator is unique. $\square$

Definition 7.2 (Canonical rival routes). *A candidate non-Black–Scholes valuation rule in the neighborhood of local derivative valuation must enter through at least one of the following routes:*

- (R1) *change the payoff identity quotient or admissible-redescription relation;*
- (R2) *use a hidden selector, calibration criterion, probability-law selector, or imported datum;*
- (R3) *enlarge the local state by volatility, rate, path, regime, inventory, funding, or collateral variables;*
- (R4) *replace payoff linearity or self-financing neutrality with nonlinear financing, preferences, transaction costs, or inventory effects;*
- (R5) *replace the terminal trace by added boundary, projection, or calibration data;*
- (R6) *replace continuous hedge-neutral local closure by a jump kernel, memory term, rough driver, higher local jet, elliptic boundary problem, hyperbolic transport law, or deterministic first-order transport;*
- (R7) *break scale covariance, time homogeneity, or the fixed deterministic risk-free numeraire.*

Theorem 7.3 (Same-scope exhaustion). *No non-Black–Scholes rival survives as a same-scope representative-invariant valuation operator in the fixed class. Every purported rival is exactly one of the following: bookkeeping-equivalent to (7.1), a declared scope change, or a representative-invariance failure.*

Proof. Let a candidate rival be non-bookkeeping-equivalent to (7.1). If it preserves all conditions in Definition 3.3, Theorem 7.1 forces it back to (7.1), contradiction. Hence it must violate at least one condition. Proposition 4.9 says that any non-bookkeeping departure occurs at one of the listed loci. Those loci are exactly the routes in Definition 7.2: payoff identity, auxiliary selector, state, financing/linearity, terminal trace, continuation law, and symmetry/numeraire. By Lemmas 2.3, 2.6, 3.2, 2.9, 3.6, 4.3, 4.5, 4.6, 4.7, 5.2, and Proposition 4.8, each route is either inert bookkeeping, an explicit change of comparison class, or a violation of quotient well-definedness. No same-scope non-Black–Scholes route remains. $\square$

Corollary 7.4 (No emergence by finite pipelines or iterative repair). *Finite compositions, alterations, calibrations, changes of measure, normalizations, or iterative procedures cannot create a new same-scope operator. If the final output is not bookkeeping-equivalent to Black–Scholes, the first valuation-bearing non-bookkeeping stage already uses one of the routes in Definition 7.2.*

Proof. Consider a finite pipeline of same-scope stages. If every stage is bookkeeping-equivalent, their composition is bookkeeping-equivalent. If the final output is not bookkeeping-equivalent, choose the first stage at which the prefix becomes non-bookkeeping. Proposition 4.9 assigns that stage to one of the departure loci, hence to a route in Definition 7.2. Iterative procedures either produce novelty at a finite stage, reducing to the same argument, or require a limit, stabilization, calibration, or completion criterion; such a criterion is imported unless it is already interface-fixed, in which case it is bookkeeping. $\square$

8 Relationship to Stochastic Representations

The theorem does not reject stochastic processes. It changes their explanatory role inside the fixed class. Once (7.1) is fixed, geometric Brownian motion, risk-neutral probability, martingale measures, and Feynman–Kac expectations are efficient coordinate representations of the same valuation operator. They preserve representative invariance when their use is a lawful representation of the already fixed quotient value. They do not by themselves supply representative invariance, because an unchosen probability law is a selector rather than a claim-invariant value; Lemma 2.8 records the precise invariance failure.

In the complete Black–Scholes market, replication removes the measure selector: all admissible representations produce the same price. In incomplete markets, no-arbitrage generally yields an interval or family of compatible prices, and an equivalent martingale measure selection is additional data. Such a selected price can be no-arbitrage-compatible while failing to be a representative-invariant value of the original one-factor claim unless the selector is made part of the market specification.

The same distinction explains implied volatility. Given an observed option price, implied volatility is the value of σ that represents the price in the Black–Scholes coordinate. It is not evidence that the underlying literally follows geometric Brownian motion with constant volatility. In the present classification, σ^2 is the residual scalar curvature coefficient left after quotient invariance, locality, hedge-neutral continuous closure, scale covariance, time homogeneity, terminal trace, and numeraire fixing have closed every other same-scope coefficient degree of freedom. If σ is allowed to become $\sigma(S, t)$, random, path-indexed, or maturity-scheduled, then the surface, factor, history, or schedule has become additional valuation data.

9 Empirical Deviations as Controlled Scope Changes

Observed deviations from constant-volatility Black–Scholes are not embarrassments for the theorem. They are diagnostics: each one identifies which fixed-class condition has been relaxed.

Table 3. Empirical deviations as controlled scope changes.

Feature or extension	Condition changed	Interpretation
Local volatility $\sigma(S, t)$	Scale covariance and/or time homogeneity	A level/time-dependent coefficient surface is added.
Stochastic volatility	One-factor locality	Variance becomes an additional state variable.
Jumps	Continuous hedge-neutral locality	A jump measure or nonlocal kernel becomes valuation-bearing.
Path-dependent derivative	One-factor locality	The state must include the path or a sufficient statistic.
Stochastic rates	Fixed deterministic numeraire	The cash account and bond curve become state-dependent market data.
Collateral or funding valuation	Self-financing invariance and payoff linearity	Implementation and balance-sheet conventions affect value.
Transaction costs	Self-financing invariance and linearity	Rebalancing is no longer neutral and valuation may become nonlinear.
Calendar regimes	Time homogeneity	Calendar labels or regime schedules enter the coefficient law.
Incomplete-market pricing measure	No hidden selector	A pricing-measure or preference selector is added.
Extra boundary condition	Terminal-trace fixation	Additional boundary data are made part of the value problem.

Thus smiles and skews do not show that the Black–Scholes operator failed inside its own fixed class. They show that observed markets require a richer class. The richer class may be empirically superior, but it is not a same-scope counterexample.

10 Related Work and Positioning

Classical Black–Scholes–Merton derivations begin with stochastic asset dynamics and derive option valuation through self-financing hedging and no-arbitrage (Black and Scholes, 1973; Merton, 1973). Classical no-arbitrage arguments establish consistency of pricing systems; the present result classifies which pricing operators remain invariant under representation-preserving transformations of a fixed claim, a strictly stronger requirement. Martingale pricing and the fundamental theorem of asset pricing are developed in Harrison–Pliska and Delbaen–Schachermayer (Harrison and Pliska, 1981, 1983; Delbaen and Schachermayer, 1994). Standard textbook treatments include Duffie, Karatzas–Shreve, Shreve, and Bjørk (Duffie, 2001; Karatzas and Shreve, 1998; Shreve, 2004; Bjørk, 2009). Change-of-numeraire methods are treated by Geman, El Karoui, and Rochet (Geman, El Karoui, and Rochet, 1995). Local-volatility, stochastic-volatility, and jump models explicitly relax pieces of the fixed class (Dupire, 1994; Heston, 1993; Merton, 1976). Standard uniqueness theory for regular parabolic terminal-value problems is available through the maximum principle and semigroup approaches (Friedman, 1964; Ethier and Kurtz, 1986).

The present paper differs in order of explanation. It does not begin by asserting a stochastic market ontology and then deriving a PDE. It begins with the conditions under which a one-factor local value is a representative-invariant value of a claim and then classifies the admissible local operator. Much of the local PDE form is standard; the contribution is the fixed-scope exhaustion showing why non-Black–Scholes alternatives necessarily change state, data, financing, boundary, symmetry, continuation law, or numeraire. Standard symmetry classifications restrict admissible equations within a modeling framework; the present argument first derives the admissible local interface from representative independence under self-financing redescription and only then applies coefficient

rigidity.

11 Conclusion

Inside the fixed one-factor local class, the Black–Scholes equation is not merely a convenient model selected from many equal same-scope options. It is the unique nondegenerate operator compatible with representative-invariant valuation, self-financing redescription, deterministic pointwise value, continuous hedge-neutral local continuation, scale covariance, time homogeneity, terminal trace, and the fixed risk-free numeraire. The residual second-order scalar is volatility: the unique same-scope curvature intensity left after the coefficient degrees of freedom have been exhausted.

The result is not an empirical claim that markets are lognormal. It is a structural claim about what remains when the valuation problem is held fixed. Empirical deviations identify which condition has been relaxed. In that sense, Black–Scholes is the representative-invariant coordinate for the fixed local class, and richer models are principled scope changes rather than same-scope refutations.

A Concrete First-Order Failure Example

Consider the first-order valuation equation

$$\partial_t V + \mu S \partial_S V = 0, \quad V(S, T) = \mathbf{1}_{\{S > K\}}, \quad (\text{A.1})$$

for a digital payoff. The solution is

$$V(S, t) = \mathbf{1}_{\{S e^{\mu(T-t)} > K\}}.$$

Replacing μ by $\mu + \varepsilon$ gives

$$V_\varepsilon(S, t) = \mathbf{1}_{\{S e^{(\mu+\varepsilon)(T-t)} > K\}}.$$

For (S, t) near the threshold, $V_\varepsilon(S, t) \neq V(S, t)$ for arbitrarily small ε . The terminal payoff has not changed; the assigned value changes because the transport coefficient has become valuation-bearing. If μ is physical drift or representative data, it is imported. If μ is fixed to risk-free deterministic transport, the rule lacks the nondegenerate quadratic curvature term. Either way, (A.1) is not a same-scope rival.

B Exhaustion Ledger

Table 4. Exhaustion ledger for attempted rivals.

Attempted rival	Required extra structure	Classification
Alternative martingale measure in incomplete market	Measure-selection rule	Hidden selector unless declared as market data.
Local volatility	Surface $\sigma(S, t)$	Scale/time scope change.
Stochastic volatility	Variance state and dynamics	State enlargement.
Jump diffusion	Jump measure or kernel	Nonlocal continuation law.
Path-dependent derivative	History or sufficient statistic	State enlargement, unless redundant and collapses to $V(S, t)$.
Transaction-cost valuation	Cost functional and nonlinear financing rule	Self-financing/linearity change.
Stochastic rates	Rate state and bond-numeraire dynamics	Numeraire/state change.
Hyperbolic or transport PDE	Characteristic data	Imported transport unless deterministic degenerate scope.
Higher-order PDE	Higher jets or additional boundary data	Hidden state or boundary selector.
Random price functional	Unresolved random value or model selector	Fails pointwise quotient representative invariance.
Extra terminal projection	Boundary representative or calibration rule	Terminal-trace scope change or selector.

No entry is classified as wrong. Each entry is classified as not being a same-scope counterexample to Theorem 7.1.

C Counterexample Checklist

A same-scope counterexample must answer all of the following affirmatively.

1. Does it preserve the same terminal-payoff quotient $\mathcal{D}_\Phi/\sim$?
2. Does it use only the one-dimensional local state (S, t) and the fixed deterministic risk-free numeraire?
3. Does it avoid hidden factors, path variables, jump kernels, volatility surfaces, calendar schedules, preference criteria, funding selectors, and pricing-measure selectors?
4. Is the assigned value pointwise deterministic at fixed (S, t) ?
5. Is it invariant under admissible hedge-accounting, coordinate, and lawful numeraire redescrptions inside the fixed class?
6. Does it preserve payoff additivity and homogeneity under frictionless self-financing portfolio composition?
7. Does it retain the same terminal trace rather than adding boundary, projection, or calibration data?
8. Does it retain continuous hedge-neutral local closure rather than a nonlocal, higher-order, hyperbolic, elliptic, jump, memory, or purely first-order continuation law?
9. Is it scale-covariant and time-homogeneous?
10. Does it produce a non-Black–Scholes operator while satisfying all of the above?

Failure at any step identifies the condition being relaxed. Passing every step returns the proposal to Theorem 7.1 and therefore to the Black–Scholes operator.

D Structural Witness Table

The following table summarizes the proof dependencies.

Table 5. Structural proof-dependency summary.

Proof layer	Mathematical work performed	Attack blocked
Maximal equivalence closure	Makes the admissible-redescription quotient canonical relative to the declared interface	Equivalence-class ambiguity.
Quotient factorization	Makes value a function on claim identity classes	Representative-dependent pricing.
Self-financing necessity	Treats capital-neutral rebalancing as claim identity rather than microstructure preference	Self-financing-as-assumption objection.
State sufficiency	Interprets locality as completeness of the one-factor state	Locality-is-imposed objection.
Selector elimination	Separates interface-fixed data from imported data	Hidden calibration, measure, or convention selectors.
Terminal-trace fixation	Fixes the European boundary data before propagation	Extra boundary or projection conditions.
Quadratic necessity and generator ceiling	Derive the second-order local operator from hedge neutralization and local continuity	Nonlocal, nonlinear, stochastic, hidden-state, first-order, or higher-order detours.
Operator normal form	Exhausts all homogeneous linear local second-order terms on the scalar state	Claims of missing local operator classes.
Scale covariance	Forces S^2 , S , constant homogeneity	Level-dependent coefficient selectors.
Time homogeneity	Makes coefficients constant	Calendar schedules and maturity-dependent laws.
Numeraire invariance/fixing	Separates lawful quotient-level numeraire changes from required pointwise coordinate fixing	Arbitrary numeraire objection.
Numeraire fixing	Uses cash and stock benchmark claims to set drift and discount	Free drift or discount choices.
Exhaustion theorem	Maps every remaining departure to a finite route family	Claims of unlisted same-scope alternatives.
Pipeline closure	Locates the first non-bookkeeping stage in any composite method	Calibration or representation changes presented as repairs.

E Dividend Adjustment

With a continuous dividend yield q , the value of holding the underlying must account for the dividend stream. The stock-replication benchmark becomes the cum-dividend self-financing relation, and the first-order coefficient is $r - q$ while the discount coefficient remains r . The equation becomes

$$\partial_t V + \frac{1}{2} \sigma^2 S^2 \partial_{SS} V + (r - q) S \partial_S V - rV = 0.$$

The representative-invariance and exhaustion arguments are unchanged; only the stock benchmark used in Lemma 6.4 is modified.

Acknowledgments

The author acknowledges the use of OpenAI ChatGPT (GPT-5.5 Thinking, accessed May 2026) for editorial assistance, journal-formatting checks, language polishing, and LaTeX revision support. The tool was used to improve presentation and compliance with journal instructions. It was not used to generate research data, images, figures, or empirical results. The author reviewed, verified, revised, and approved all AI-assisted output and takes full responsibility for the originality, validity, and integrity of the manuscript.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Disclosure of interest

The author reports no competing interests to declare.

Data availability statement

No datasets were generated or analyzed during the current study.

CRedit author statement

Amos Jay Maley: Conceptualization; Formal analysis; Methodology; Writing – original draft; Writing – review and editing.

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